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CECS FINAL PROJECT - FALL 2023

Group 2 - Final Report

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Chapter 1: Introduction

1.1 Project Background

Water is an essential element for life on the planet Earth. It serves as a lifeline for communities, essential for the daily usage of not only humans but also other species. However, some water body especially rivers and lakes, where waters from locals and factories flow straight in to might be harmful, posing significant health risks for citizens who live near those sources.

- Exposure to polluted water can lead to many health problems, including gastrointestinal diseases, skin infections, and respiratory problems.
- Polluted water sources can contain toxic chemicals, heavy metals and pathogens, which cause long-term health effects such as cancer, organ damage and developmental disorders.
- Agricultural activities relying on polluted water for can result in illness in crops, causing risks to food safety and public health.
- Industries dependent on water resources may face serious damage and financial losses
- Polluted water can contaminate drinking water sources, access drinking water will no longer be safe, particularly in rural areas with limited access to alternative water sources.

Take the Citarum River in Indonesia as an example. However, years of uncontrolled industrial, wastewater and household waste have turned the river into one of the world's most polluted rivers. The impact on human life is negative in multiple aspects

- First, pollution has led to a public health crisis, causing people to suffer from water-borne diseases such as cholera, dysentery, and skin infections due to exposure to contaminated water.
- A lack of clean water for drinking and sanitation perpetuates vicious cycles of disease and poverty, especially affecting vulnerable populations such as children and the elderly.
- The decline in fish populations due to pollution has devastated local fishing communities, robbing them of their livelihoods and traditional lifestyles. Polluted river water also hinders agricultural productivity because farmers rely on it for irrigation, leading to crop failures and economic hardship.

- The growing tourism industry along the Citarum almost disappeared as tourists were deterred by unpleasant odors and pollution, further exacerbating the economic downturn in the area.
- The social fabric of communities along the river has been shaken as people grapple with losing their health, livelihoods and cultural heritage.

Therefore, to have an effective way to manage the water quality, not only for early warning about dangerous and harmful water sources but also to educate and raise people's awareness about keeping the clean environment, especially lakes and rivers

We aim to develop a convenient and accessible tool for the public to acknowledge the water quality with a broader view. It can provide citizens with valuable statistics and data on the condition of water resources within their localities and other areas. By aggregating data from diverse sources, including local water quality reports, scientific studies, and community-driven surveys, we want to construct an accurate water quality map across different regions.

1.2 Project Definition

Our project named Water Monitoring Live Map will use specialized hardware to collect valuable water quality data from interested water bodies, which then will be displayed in a real-time online map. Its functions will follow that of many existing air quality maps but employ many features and designs tailored toward water instead. This innovative tool will provide consumers and regulatory entities immediate access to critical water quality information in pursuit of improving public health and safety. The Water Monitoring Live Map is an online platform emphasizes accessibility, offering real-time visualization and analysis of water quality data. Its goal is to provide easy access to important information about the health and condition of the water bodies, offering data into various safety parameters, and giving early warning to the locals in case of emergency.

1.3 Project Objectives

Our project, named Water Monitoring Live Map, is going to be a new, modern and overall view of how we monitor water quality in lakes/rivers/ and other water sources. Inspired by existing air quality maps, our online platform will

- Display real time data collected from sensors placed across various points of a water body.
- Provide statistical data into important water quality parameters such as pH levels, dissolved oxygen content, turbidity, and pollutant concentrations.
- Optimize algorithms and sensor technology to continuously update the map, and provide accurate and up-to-date information on the pollution level of water sources.
- Feature user-friendly interface websites, allowing both the general public and the governments to access and interpret the data.

- Enable users to visualize water quality trends over time by using graphs, identify potential threats and early warning about any unusual data.

By increasing transparency, our project aims to provide communities with a way to monitor and protect their water resources. Water Monitoring Live Map serves as a valuable tool to promote public health and environmental awareness. While achieving full automation may still be ongoing, our team is committed to using the most modern technology and innovation to solve pollution challenges and create a positive impact on society. We believe that realizing our vision of a healthier, more sustainable future is not just an impossible goal but an achievable reality.

1.4 Project Specifications

I. System Description

The entire system consists of 2 parts: Water Stations (Sensor) and a server. The water stations have to be placed in deep water, which is used all around (24/7), even in rough weather. The operation of each water station depends on batteries, and each should last for more than 3 months before the next charge (maintenance). The real-time quality map will be on 24/7 and will be linked with a database to store and retrieve data.

1.4.1. A floating platform with weights and wire to fix the whole system in one place consisting of:

- a. Floating Platform
 - Rubber float
- b. Underwater fixing system:
 - Wire
 - Elastic Wire
 - Weight

1.4.2. An integrated sensor system consisting of

- a. Measurement Instruments:
 - Thermometer
 - pH sensor
 - Dissolved oxygen meter
 - TDS meter
 - EC meter

1.4.3. A Control System to manage sensors, visualize working states of the station and send data back to the server consisting of:

- a. Sensor Data Collecting and Process

- A MCU
- b. Data wireless transfer system
 - Central router
 - Wifi-enabled chips
- c. Control system support
 - Power supply system
- d. Working state indication system
 - RGB LED
 - One operator monitor
 - One operator console

1.4.4. A server to receive data and process the data consisting of:

- a. Water station management system
 - Connection and water station state checker
 - Water station position approximate
- b. Plotting and graphing system
 - Automatic data-pushing system
 - Automatic mapping data to the water quality map
 - Real-time interactive map
 - History database
 - Comparison section
 - Advice from experts on water treatments
- c. Data recording system
- d. User management system
 - Feedback/Report

II. Purpose and Operation

2.1. Use Scenario

A water monitoring system plays a critical role in assessing the cleanliness and quality of water in various locations. Particularly essential in remote or inaccessible areas where direct measurements are challenging, this system operates autonomously to monitor water quality over extended periods without human intervention continuously. By consistently updating information to a centralized database, it ensures that real-time data on water quality is readily available for access. This approach enables individuals and organizations to effortlessly retrieve pertinent information about water cleanliness in specific areas whenever necessary. Through the implementation of such a system, we can effectively keep track of water quality across different locations, particularly in regions that are difficult to reach or require ongoing monitoring to maintain environmental standards and public health.

2.2. Assumptions, Constraints, and Dependencies

The entire system is designed to function autonomously, eliminating the need for any operator intervention. It operates seamlessly by constantly sending data to the database via WiFi connectivity. Additionally, it's programmed to alert the server when the battery levels are low, ensuring uninterrupted operation. The system maintains a continuous connection to both the server and the website, ensuring that data is always accessible. Moreover, it is equipped with sensors to detect any decline in water quality below the standard thresholds, promptly sending warnings to the appropriate recipients. This comprehensive approach ensures that the water monitoring system operates efficiently and effectively, providing reliable data and alerts to maintain water quality standards.

III. Sensor Block Requirements

3.1. Functional requirements

3.1.1. Requirements for Controller Unit:

- The Controller Unit should be as most energy efficient as possible
- The Controller Unit should be able to read data from any sensor and transfer data via standardized communication methods like:
 - + Wireless
 - + Wired
- The Controller Unit size should be small and compact
- The Controller Unit should be able to sample data at a fast pace with great accuracy and precision
- The Controller Unit should run fast, and precise even during heavy-duty tasks
- The Controller Unit should be able to integrate into any circuits
- The Controller Unit should not be affected by any environmental conditions
- The Controller Unit should be able to check and show its:
 - + Working state
 - + Sensor working state
 - + Connection state
 - + Battery Level
- The Controller Unit should have basic MCU functional blocks including:
 - + Peripheral
 - + Oscillator
 - + Memory
 - + Microprocessor Unit
 - + GPIO pins
- The Controller Unit should not be affected by any signal noises during operation
- The Controller Unit should be able to take sensor measurements and send them back to the server after a fixed amount of time

- The Controller Unit should be put on a high-quality PCB

3.1.2. Requirements for Wireless Communication Modules

- The WCM should transfer data reliably over any weather condition
- The WCM should be able to transfer data to every router
- The WCM size should be small and compact
- The WCM should be an onboard extension of the MCU

3.1.3. Requirements for power sources

- The power supply system shall be an independent power source – an internal battery.
- The power supply system shall be resistant to
 - + The short circuit
 - + The reverse polarity of the power supply voltage
 - + The power supply voltage drops down to 0V
 - + The overcharging, the overheating, and the deep discharge of internal batteries
- The power supply system shall provide an audible alarm:
 - + When each 10% battery drops.
 - + 3 hours before reaching the maximum permissible discharge internal batteries level.
 - + 1 hour after reaching the maximum permissible charging level.
- The power supply system shall provide information about the current state of power sources i.e. voltage, current charge/discharge, power consumption, the estimated capacity of internal batteries, and estimated working time of internal batteries.
- The length of time for a full charge of the batteries must not exceed 12 hours.

3.1.4. Requirements for sensors

- Accurate and precise in measuring all relevant metrics
- Resistance to the underwater environment
- Short update cycle to maintain real-time measurement capabilities.

3.1.5. Requirements for the container

- Waterproof to protect internal components
- Aerodynamic shape to reduce the drag force of the river tides

3.1.6. Requirements for the float

- The buoyant force must be enough to lift the battery
- Have the roof cover to protect the battery

3.1.7. Requirements for the anchor

- Heavy enough to nail the station in normal river tide condition

- Heavy enough to stand against strong wind/storm, water current

3.1.8. Requirements for the wire

- Hollow in the center to carry electric wire
- Need multiple layers for multiple functions
 - + Electric power transmission
 - + Water protection
 - + Stand against the water pressure
 - + Flexible for movement underwater

3.1.9. Requirements for the chain

- Resistant the underwater environment
- 1 chain connects the sensor to the anchor and 3 chains connect the sensor to the float

3.2. Non-Functional Requirements

3.2.1. Performance

a. Controller Unit

- ADC resolution of at least 12-bit
- Processor Clock Speed at least 60MHz and maximum 240MHz
- AHB
- 32-bit Xtensa Instruction Set Architecture
- At least 4MB of Flash memory
- At least 512 KB SRAM memory
- At least 384 KB ROM
- FreeRTOS support
- Standardized wired data transmission methods like I2C, SPI, I2S, DAC, Serial
- Standardized wireless data transmission methods like TCP/IP
- Energy efficient utilizing:
 - + ULP
 - + Deep sleep whenever not taking sensor measurements or sending data back to the server
 - + RTC for keeping track of time
 - + 3.3v logic level
 - + PMU

b. Wireless communication modules

- RF Receiver:
 - + CMOS power amplifier
 - + 2.4 GHz
 - + 802.11 b/g/n

- + 150Mbps of data rate
- RF Transmitter:
 - + AGC
 - + High-resolution, High-speed ADCs
 - + Up to 20.5dBm of transmitting power
- c. Power source
 - Must be able to power the system for at least 3 months
 - Approximately 3.7V 4.2V
 - Capacity 20.000mAh
 - Material: Lithium Ion
- d. Sensors
 - The sensor should provide water quality measurements with an accuracy of $\pm 2\%$ for critical parameters such as pH, dissolved oxygen, and pollutant concentrations
 - The sensor should have a rapid response time, providing real-time or near-real-time measurements to capture sudden changes in water quality.
 - The sensor should be capable of transmitting data to a central system or database using industry-standard communication protocols
 - The sensor should have efficient power consumption to enable long-term deployments
- e. The container
 - Material: Nanocomposite
- f. Float
 - Density: Must be less than 1 gram/cm^3
 - Material: High-density polyethylene (HDPE)
 - Load: 5 kg load buoyancy
- g. Metal anchor
 - Density: $>1000 \text{ kg/m}^3$
 - Material: AISI 304 Stainless Steel (UNS S30400)
 - Dimension: mushroom anchor, 25cm in diameter for the cap (top part) and 30 cm in height
 - Weight: 5 kg
- h. Wire:
 - 4 Layers from innermost to outermost:
 - + Conductor: Copper
 - + Insulator: PVC
 - + Mechanical protection (armors): Aluminum
 - + Outer sheath: Polyurethane (PU)
- i. Cable:
 - Material: AISI 304 Stainless Steel (UNS S30400)

3.2.2. Reliability

- a. Controller Unit
 - Faraday cage closing
 - Cryptographic Hardware Acceleration:
 - + SHA
 - + RSA
 - + AES
 - + RNG
 - Secure Boost
 - Flash Encryption
 - Hardware debugging
- b. Wireless communication modules
 - RF Transmitter:
 - + Calibration routines for radio imperfections
 - RF Receiver:
 - + RF filter
 - + Baseband filter
 - + DC offset cancelation circuits
 - Extension antenna
 - WPA3 encryption
- c. Battery
 - The capacity reduces by less than 10% after 1 year of functioning.
 - Function normally in moderately hotter/colder temperature
 - The battery should provide a reliable and consistent power supply over its lifespan.
- d. Sensors
 - Data accuracy stays consistent after at least 1 year
- e. The container
 - Keep the internal component waterproof even in hazardous weather conditions
- f. Float
 - Maintain buoyancy in harsh conditions such as storms, high waves, and attacks from wildlife.
- g. Metal anchor
 - The combination of weight and shape allows the anchor to sink into the soft bottom of rivers
- h. Wire
 - Maintain insulation with the environment under any scenario
- i. Cable

- Sufficient length to allow the device a circular motion with a 2-meter diameter circle

3.2.3. Durability

- a. Controller Unit
 - At least 10,000 program/ erase Flash memory cycles
 - LDO (5v 3.3v) for power supply/ logic voltage protection
 - High-quality components
 - Capacitor for reducing power supply noise and spike
 - The life expectancy of the controller unit should not be less than 20 years
 - The whole controller unit should come on a PCB that has:
 - + At least 3 layers
- b. Wireless communication modules
 - Plastic enclosing
 - Onboard antenna
- c. Battery
 - Does not cause vital damage to the system in case of malfunction
 - It must withstand environmental factors such as temperature variations, humidity, and vibrations without significant degradation.
- d. Sensors
 - Automatically disable when malfunction, avoid chain damage to other components
- e. The container
 - Can stand against the water pressure for a long time
 - Absorb friction and keep the internal components stable
- f. Float:
 - Puncture resistance: 200 Newtons (N)
- g. Metal anchor:
 - Tensile strength: 515 MPa
 - Yield strength: 205 MPa
 - Elongation in 50 mm: 50%
 - Hardness: 80 HRC
- h. Wire:
 - Resistance to external environmental factors
 - Resistance against fish attacks
- i. Cable:
 - Working Load Limit: 200 N/mm^2
 - Tensile strength: 710 MPa

3.2.4. Serviceability

- a. Controller Unit
 - Can work normally in the range of -20C to 85C
 - The Controller Unit should come from a manufacturer with library and SDK support for programming the Controller Unit
- b. Wireless communication modules
 - Support antenna diversity
 - The PCB should pass the IPC-2221 standard
 - Follow wireless communication methods standards such as:
 - + TCP/IP
 - + SoftAP
 - + Station Mode
- c. Battery
 - Easy to recharge/replace, ease of maintenance and repair, ensuring that the container can be serviced efficiently without compromising its functionality.
- d. Sensors
- e. The container
 - Easy to connect/disconnect with cables
- f. Float:
 - Mendable for minor damage instead of full replacement
- g. Metal anchor:
 - Can undergo chemical cleaning with minimal damage
- h. Wire:
 - Designed with flexible material, allowing for restoration of connectivity in the event of being severed or cut
- i. Cable:
 - Can undergo chemical cleaning with minimal damage
 - Chain links can be individually replaced without harming the total structure

3.2.5. Conformance

- a. Controller Unit
 - 32-bit Xtensa instruction set architecture
 - Passed the following standards:
 - + EMC
 - + FCC
 - + CE
 - + RoHS
- b. Wireless communication modules
 - Pass the following standards:
 - + 802.11 b/g/n standard
 - + FCC

- + CE
- + IC
- + TELEC
- + RoHS

c. Battery

- The battery should meet safety standards to prevent overheating, leakage, or other potential hazards.
- Compliance with international standards such as IEC 62133 (for rechargeable batteries) or relevant regional safety certifications is required.
- The battery should comply with environmental regulations, such as RoHS

d. Sensors

- Following ISO 22013 Marine Environment Sensor performance

e. The container

- The container must conform to material standards ensuring durability, safety, and environmental sustainability

f. Float

- Following ISO 12402 Personal flotation devices

g. Metal anchor

- Following ISO 15385 for "General principles for design Anchors."

h. Wire

- Following ICPC Recommendations such as issue 1-14A, 2-12C, 12-2F, and 19-1
- IP 68 standard

i. Chains

- Following the American Bureau of Shipping (ABS) standard for mooring chains

3.2.6. Aesthetics

a. Controller Unit

- Controller Unit Dimension: 25.5mm x 18mm x 3.1mm
- Whole PCB dimension: 8cm x 8cm x 3mm
- SMD packed

b. Wireless communication modules

- SMD packed
- Dimension should not surpass 10mm x 10mm x 3mm

c.

- Fit perfectly inside the float with enough room for thermal

d. Sensors

- Dimensions: Fit perfectly inside the container with enough room for thermal expansion due to changes in environmental temperature

e. The container

- Dimension: 20x20x20 cm

- Shape: Chrysalis
- f. Float
 - Dimension: 1 m in diameter 20 cm in height
 - Bright color allows easier collection
- g. Metal anchor
- h. Wire
 - Length: 5-7 m (shorter than the chains)
 - 8 mm in diameter
 - Bright color so people can easily identify it to be an electrical wire
- i. Cables
 - Length: Minimum 7 m
 - Dimension: 8 mm in diameter for each chain link

3.2.7. Others

- a. Controller Unit
 - The whole controller unit should be able to make decisions on its own without any operator
 - The cost should not be higher than 30 USD per controller unit
- b. Wireless communication modules
 - The cost for each module should be less than 5 USD
- c. Battery
 - The cost per battery unit should be within a specified budgetary limit.
- d. Sensors
 - The cost per sensor unit should be within a specified budgetary limit.
 - Consideration should be given to achieving a balance between cost and sensor performance
- e. The container
 - The cost for each module should be less than 5 USD
- f. Float
 - Utilize multiple connection points to distribute the strain evenly, thereby mitigating the concentration of stress on a single location.
- g. Metal anchor
 - Should be able to handle, deploy, and retrieve without complexities or risks during operation.
- h. Wire
 - Implement in a way to minimize straining
- i. Chains
 - Smartly implemented to avoid entanglement.

IV. Database Requirements

4.1. Functional Requirements

4.1.1. Requirements for user interface

- Display data from all sensors.
- Update regularly after each 2-hour period.
- Provide an interactive environment and allow users to send reports and feedback for specific places.
- Have different layers to display different characteristics of water quality such as pH, temperature, turbidity, etc.
- All layers shall have details displayed on a default map corresponding to a scale of 1:1000 for Vietnam.
- Provide users with basic solutions to prevent using low-quality water.
- Provide interactive map functions, such as zooming, panning, rotating, measuring, etc., and responding to the user's input and gestures.
- Real-time map marked with specific colors in specific areas.

4.1.2. Requirements for the server

- Load and display the map data from various sources and formats, such as shapefiles, GeoJSON, KML, etc.
- Integrate with external web services and APIs, such as geocoding, geolocation, geofencing, etc., and access and use their data and functions.
- Handle a lot of concurrent and distributed requests from multiple users and devices.
- provide authentication and authorization mechanisms for the map users and developers and control the access and permissions of the map data and functions.
- Export and import the map data in different formats and methods, such as downloading, uploading, printing, sharing, etc.
- Provide search and query functions for the map data and features and allow the user to filter, sort, and select the map elements based on different criteria and conditions.
- Adapt to different browsers and devices and ensure the compatibility and usability of the map application and its features.

4.1.3. Requirements for the database

- Support data security and privacy functions.
- Protect the data and the user information from unauthorized access, modification, or deletion, and prevent cyberattacks such as hacking, malware, or denial-of-service.
- Support data backup and recovery functions, and create and restore backups of the data, and prevent data loss or corruption in case of emergencies or disasters.

- Store and organize the water quality data from other various sources, such as USGS NWIS and BioData, EPA STORET, and USDA-ARS STEWARDS1, GEMStat2, WISE Water Framework Directive Quality Elements3, etc.
- Support data mining and discovery functions for the water quality data.
- Support data monitoring and evaluation functions, and have feedback and recommendations for improvement.

4.2. Non-Functional Requirements

4.2.1. Performance

- The system needs to constantly update the data once a day
- The website should load and display maps with water quality data within 3 seconds of a user's request.
- The system should support a minimum of 100 concurrent users without significant degradation in performance.

4.2.2. Reliability

- The website do not promote/accept advertisement, keep it non-profit
- The data must be up-to-date and precise
- The website should have an uptime of at least 99.9%.
- The system should employ encryption (e.g., HTTPS) to secure data transmission between the server and clients

4.2.3. Durability

- The server and website have a backup cloud version in case of restoration
- The server must be able to withstand cyber attacks with small size (e.g DDoS of 10 gigabits per second
- The backend of the website must be easy to understand, rewritable, easy to upgrade, and easy to add modules
- Regular updates and patches should be applied to address security vulnerabilities and improve system functionality.

4.2.4. Serviceability

- Having backup data displayed when the server got Ddos.
- Easy to restore from the backup version
- Accessibility standards (e.g., WCAG) should be followed to ensure the website is usable by individuals with disabilities.
- The website should be compatible with major web browsers (e.g., Chrome, Firefox, Safari, Edge) and different devices (desktops, tablets, and smartphones).
- Compatibility with various screen sizes and resolutions should be considered

4.2.5. Conformance

- Comply with Vietnamese law
- Must be deployed with no image or contents that violate intellectual property law
- The system should comply with relevant regulations and standards related to water quality data, privacy, and web application security.
- Any third-party mapping APIs or data sources should be used in compliance with their respective terms of use

4.2.6. Aesthetic

- User-friendly and minimal interface.
- Real-time maps must be marked with specific colors in specific areas.

4.2.7. Other

- Cost: 10\$ hosting fee each year.

V. Logistics, Non-Functional Requirements

5.1 Documentation Requirements

5.1.1 Manual including:

- Functional description of the equipment
- Detail technical parameters
- Description of installation and activation procedures
- Description of assembly and disassembly
- Operational and maintenance manuals
- Procedure to deal with the system if any error like the sensor disconnects from the server or the water sensor of the sensor block stops working,...

5.1.2 Maintenance documentation shall include:

- Mechanical drawings
- Part lists for all circuits
- Wire diagram that shows the interconnection of modules, boards

5.2 Maintenance Requirements

- The whole system should be checked once every 3 months for battery chargings, sensor maintenance
- The server and the water quality map should be maintained every day to keep the server from going down or losing connection with the sensors.

5.3 Verification

- The sensor system should come with verification and a result of successful laboratory verification of the system's working condition deep underwater, the accuracy reading of the sensor, the stable connection between the sensors and the server, and the stable data transfer. The certification shall be issued by a laboratory accredited for tests required by an accrediting authority of any of the countries

5.4 IP and software copyright

- Any IP or software should have appropriate licenses

Chapter 2: Project Management

2.1 Project Plan

No	Task ID	Priority	Status	Task Name	Description	Duration	Start Date	End Date	Deliverables	Assigned to	Resource	Predecessors
1	CECS1001	High	Completed	Create functional requirements for the sensor block	Create specifications for the sensor block	1 Week	30/11/2023	7/12/2023	A document on the requirements	Khang	The operator and the customer	None
2	CECS1002	High	Completed	Create functional requirements for the float and wires block	Create specifications for the float and wires block	1 Week	30/11/2023	7/12/2023	A document on the requirements	Trung	The operator and the customer	None
3	CECS1003	High	Completed	Create functional requirements controller block	Create specifications for the controller block	1 Week	30/11/2023	7/12/2023	A document on the requirements	Quý	The operator and the customer	None
4	CECS1004	High	Completed	Create functional requirements for the map UI/UX	Create specifications for the UI/UX	1 Week	30/11/2023	7/12/2023	A document on the requirements	Thư	The operator and the customer	None
5	CECS1005	High	Completed	Create functional requirements for the backend of the map	Create specifications for the backend of the map	1 Week	30/11/2023	7/12/2023	A document on the requirements	Cường	The operator and the customer	None
6	CECS1006	High	Completed	Create a final functional requirements for the whole system	Create a table of functional requirements for the whole system	3 Days	7/12/2023	10/12/2023	A document on the requirements	Quý	The operator and the customer	None
7	CECS1007	High	Completed	Create non-functional requirements for the sensor block	Create specifications for the sensor block	1 Week	10/12/2023	17/12/2023	A document on the requirements	Khang	The operator and the customer	CECS1001
8	CECS1008	High	Completed	Create non-functional requirements for the float and wires block	Create specifications for the float and wires block	1 Week	10/12/2023	17/12/2023	A document on the requirements	Trung	The operator and the customer	CECS1002
9	CECS1009	High	Completed	Create non-functional requirements controller block	Create specifications for the controller block	1 Week	10/12/2023	17/12/2023	A document on the requirements	Quý	The operator and the customer	CECS1003
10	CECS1010	High	Completed	Create non-functional requirements for the map UI/UX	Create specifications for the UI/UX	1 Week	10/12/2023	17/12/2023	A document on the requirements	Thư	The operator and the customer	CECS1004
11	CECS1011	High	Completed	Create non-functional requirements for the backend of the map	Create specifications for the backend of the map	1 Week	10/12/2023	17/12/2023	A document on the requirements	Cường	The operator and the customer	CECS1005
12	CECS1012	High	Completed	Create a final non-functional requirements for the whole system	Create a table of non-functional requirements for the whole system	3 Days	17/12/2023	20/12/2023	A document on the requirements	Quý	The operator and the customer	CECS1006
13	CECS1013	High	Completed	Create a plan	Make a main plan for the product development	3 Days	20/12/2023	23/12/2023	An outline for the plan in table format	Quý	The operator	All task above
14	CECS1014	Medium	Completed	Create a task table for the team members	Split the task and assigned it to different members	3 Days	23/12/2023	26/12/2023	An outline for the task table	Quý	The operator	CECS1013
15	CECS1015	Medium	Completed	Create backup plans	Make backup plans in case the main plan failed	2 Days	26/12/2023	28/12/2023	An outline for the backup plan in table format	Trung	The operator	CECS1013
16	CECS1016	Medium	Completed	Analyze the requirements	Analyze the requirements for each block	1 Week	28/12/2023	4/1/2024	A table on requirements of each block	Cường	The operator and the customer	All task above
17	CECS1017	Medium	Completed	Design high level block diagram	Create a high level workflow of the product	2 Days	4/1/2024	6/1/2024	A graph on the workflow of the whole system	Khang	The operator and a computer	All task above
18	CECS1018	Medium	Completed	Design block diagram for each component	Create the workflow for each block	5 Days	6/1/2024	11/1/2024	A graph on the workflow of the each block	Thư	The operator and a computer	CECS1017
19	CECS1019	Medium	Completed	Design sensor block	Decide what components would be in the block and the combination of the components	5 Days	11/1/2024	16/1/2024	A raw design of the block	Khang	The operator and a computer	CECS1017
20	CECS1020	Medium	Completed	Design casing block	Decide on the shape of the case	5 Days	11/1/2024	16/1/2024	A raw design of the block	Trung	The operator and a computer	CECS1017
21	CECS1021	Medium	Completed	Design float and wires block	Decide the shape of the float and the number of layers on each wire	5 Days	11/1/2024	16/1/2024	A raw design of the block	Trung	The operator and a computer	CECS1017
22	CECS1022	Medium	Completed	Design and program the controller block	Design the programming architecture of the controller	5 Days	11/1/2024	16/1/2024	A raw design of the block	Cường	The operator and a computer	CECS1017
23	CECS1023	Medium	Completed	Design website frontend block	Decide the website look	5 Days	11/1/2024	16/1/2024	A raw design of the block	Quý	The operator and a computer	CECS1017
24	CECS1024	Medium	Completed	Design website backend block	Decide the server data management and communication protocol	5 Days	11/1/2024	16/1/2024	A raw design of the block	Thư	The operator and a computer	CECS1017
25	CECS1025	Medium	Completed	Test sensor block	Test if the sensor first design pass the requirements	2 Days	16/1/2024	18/1/2024	A document on the qualification of the block	Thư	The operator and a computer	All task above
26	CECS1026	Medium	Completed	Test casing block	Test if the case first design pass the requirements	2 Days	16/1/2024	18/1/2024	A document on the qualification of the block	Trung	The operator and a computer	All task above
27	CECS1027	Medium	Completed	Test float and wires block	Test if the float and wires first design pass the requirements	2 Days	16/1/2024	18/1/2024	A document on the qualification of the block	Trung	The operator and a computer	All task above
28	CECS1028	Medium	Completed	Test the controller block	Test if the controller functionalities pass the requirements	2 Days	16/1/2024	18/1/2024	A document on the qualification of the block	Quý	The operator and a computer	All task above
29	CECS1029	Medium	Completed	Test web frontend block	Test if the website frontend pass the requirements	2 Days	16/1/2024	18/1/2024	A document on the qualification of the block	Thư	The operator and a computer	All task above
30	CECS1030	Medium	Completed	Test web backend block	Test if the website backend pass the requirements	2 Days	16/1/2024	18/1/2024	A document on the qualification of the block	Thư	The operator and a computer	All task above
31	CECS1031	Medium	Completed	Design alternative for the unqualified blocks	Design alternative for the unqualified blocks	1 Week	18/1/2024	24/1/2024	A document on the qualification of the block	Quý	The operator and a computer	All task above
32	CECS1032	Medium	Completed	Select the best alternatives	Select the best alternatives	1 Day	24/1/2024	25/1/2024	A document on selected alternative	Quý	The operator and a computer	All task above
33	CECS1033	Medium	Completed	Test the sensor block with simulation	Test the sensor on computer in working condition simulation	1 Day	25/1/2024	26/1/2024	A document on the qualification of the block	Khang	The operator and a computer	All task above
34	CECS1034	Medium	Completed	Test the casing block with simulation	Test the case on computer in working condition simulation	1 Day	25/1/2024	26/1/2024	A document on the qualification of the block	Trung	The operator and a computer	All task above
35	CECS1035	Medium	Completed	Test the float and wires block with simulation	Test the float and wires on computer in working condition simulation	1 Day	25/1/2024	26/1/2024	A document on the qualification of the block	Trung	The operator and a computer	All task above
36	CECS1036	Medium	Completed	Test the web backend security and privacy mechanism with simulation	Test the web backend security and privacy mechanism on computer in working condition simulation	1 Day	25/1/2024	26/1/2024	A document on the qualification of the block	Thư	The operator and a computer	All task above
37	CECS1037	Medium	Completed	Test the web data and user management with simulation	Test the web data and user management on computer in working condition simulation	1 Day	25/1/2024	26/1/2024	A document on the qualification of the block	Cường	The operator and a computer	All task above
38	CECS1038	Medium	Completed	Test web flow and UI/UX with simulation	Test the web flow and UI/UX on computer in working condition simulation	1 Day	25/1/2024	26/1/2024	A document on the qualification of the block	Thư	The operator and a computer	All task above
39	CECS1039	Medium	Completed	Order sensor block components	Find the best provider and purchase sensors	2 Days	26/1/2024	28/1/2024	A document on the purchase of sensor and the physical sensors	Quý	The operator and a component provider	All task above
40	CECS1040	Medium	Completed	Order casing and float/ wires block material	Find the best provider and purchase necessary material	2 Days	26/1/2024	28/1/2024	A document on the purchase of material and the physical material	Trung	The operator and a material provider	All task above
41	CECS1041	Medium	Completed	Purchase web hosting and database services	Find the best provider and purchase web hosting and database services	2 Days	26/1/2024	28/1/2024	A document on the purchase of material and the software	Thư	The operator and a software provider	All task above
42	CECS1042	Medium	Completed	Make sensor block prototype	Build realife prototype of the sensors	5 Days	28/1/2024	2/2/2024	A document on the qualification of the block	Khang	The physical sensor	CECS1039
43	CECS1043	Low	Completed	Make casing block prototype	Build realife prototype of the case	5 Days	28/1/2024	2/2/2024	A document on the qualification of the block	Trung	The physical case	CECS1040
44	CECS1044	Low	Completed	Make float and wires block prototype	Build realife prototype of the float and wires	5 Days	28/1/2024	2/2/2024	A document on the qualification of the block	Trung	The physical float and wires	CECS1040
45	CECS1045	Low	Completed	Make website prototype	Build realife prototype of the website	5 Days	28/1/2024	2/2/2024	A document on the qualification of the block	Thư	The website	CECS1041
46	CECS1046	Low	Completed	Make database hierarchy prototype	Build the prototype of database workflow and data management	5 Days	28/1/2024	2/2/2024	A document on the qualification of the block	Cường	The database	CECS1041
47	CECS1047	Low	Completed	Make microcontroller prototype firmware	Build user-friendly firmware with reliable functionalities	5 Days	28/1/2024	2/2/2024	A document on the qualification of the block	Quý	The microcontroller	All task above
48	CECS1048	Low	Completed	Test the sensor block in working condition	Test the sensor in realife working condition	2 Days	2/2/2024	4/2/2024	A document on the qualification of the block	Khang	The operator and a test place	CECS1042
49	CECS1049	Low	Completed	Test the float and wires block in working condition	Test the float and wires in realife working condition	2 Days	2/2/2024	4/2/2024	A document on the qualification of the block	Trung	The operator and a test place	CECS1043

50	CECS1050	Low	Completed	Test the website in deployment condition	Test DDos attack, data breach and data leak	2 Days	2/2/2024	4/2/2024	A document on the qualification of the block	Cường	The operator and a test place	CECS1045
51	CECS1051	Low	Completed	Manufacture the float	Manufacture the float	1 Week	4/2/2024	11/2/2024	A final physical float	Trung	A manufacturer	All task above
52	CECS1052	Low	Completed	Manufacture the wires	Manufacture the wires	1 Week	4/2/2024	11/2/2024	Final physical wires	Trung	A manufacturer	All task above
53	CECS1053	Low	Completed	Manufacture the PCB and the control unit	Manufacture the PCB and the control unit	1 Week	4/2/2024	11/2/2024	Final PCB and controller unit	Quý	A manufacturer	All task above
54	CECS1054	Low	Completed	Create a final water quality map	Finalize the water quality map features	1 Day	11/2/2024	12/2/2024	Final water quality map website	Thư	The operator and software tools	All task above
55	CECS1055	Low	Completed	Deploy the product	Deploy the product	1 Day	12/2/2024	13/2/2024	A deployable product	Quý	A marketing strategy	All task above
56	CECS1056	Low	Completed	Write user manual	Write user manual	1 Day	13/2/2024	14/2/2024	An instruction document for the customer	Quý	The operator and a document	All task above
57	CECS1057	Low	Completed	Write maintenance notes	Write maintenance notes	1 Day	14/2/2024	15/2/2024	An instruction document for the customer	Quý	The operator and a document	All task above
58	CECS1058	Low	Completed	Write emergency case handle	Write emergency case handle	1 Day	15/2/2024	16/2/2024	An instruction document for the customer	Quý	The operator and a document	All task above
59	CECS1059	Low	Completed	Present and demo the system	Present and demo the system	1 Day	16/2/2024	17/2/2024	Work and reliable system	Quý	The operator and a completed system	All task above

[Table 1. Project plan and distribution](#)

2.2 Contribution of Team Members

Task name	Dương Nghiệp Quý Electrical Engineering	Lê Hải Trung Mechanical Engineering	Hoàng Anh Thư Computer Science	Nguyễn Quốc Cường Computer Science	Trần Minh Khang Computer Science
Create functional requirements for the sensor block	R	A	I	I	I
Create functional requirements for the float and wires block	C	R	I	C	I
Create functional requirements controller block	R	R	I	I	I
Create functional requirements for the map UI/UX	C	R	I	I	I
Create functional requirements for the backend of the map	C	C	A	R	R
Create a final functional requirements for the whole system	I	I	R	R	A
Create non-functional requirements for the sensor block	C	I	A	C	R
Create non-functional requirements for the float and wires block	I	C	C	A	C
Create non-functional requirements controller block	I	I	A	R	R
Create non-functional requirements for the map UI/UX	C	R	I	A	I

Create non-functional requirements for the backend of the map	R	R	A	R	C
Create a final non-functional requirements for the whole system	R	R	C	A	I
Create a plan	C	C	A	R	R
Create a task table for the team members	I	I	R	R	A
Create backup plans	C	I	A	C	R
Analyze the requirements	I	C	C	A	C
Design high level block diagram	I	I	A	C	R
Design block diagram for each component	A	R	R	R	C
Design sensor block	R	A	I	I	I
Design casing block	C	R	I	I	I
Design float and wires block	R	R	I	C	I
Design and program the controller block	C	R	I	I	I
Design website frontend block	C	C	A	R	R
Design website backend block	I	I	R	R	A
Test sensor block	C	I	A	C	R
Test casing block	I	C	C	A	C
Test float and wires block	I	I	A	R	R
Test the controller block	C	R	I	A	I
Test web frontend block	R	R	A	R	C
Test web backend block	R	R	C	A	I
Design alternative for the unqualified blocks	C	C	A	R	R
Select the best alternatives	I	I	R	R	A
Test the sensor block with simulation	C	I	A	C	R
Test the casing block with simulation	I	C	C	A	C
Test the float and wires block	I	I	A	C	R

with simulation					
Test the web backend security and privacy mechanism with simulation	I	C	C	A	C
Test the web data and user managment with simulation	I	I	A	C	R
Test web flow and UI/UX with simulation	A	R	R	R	C

[Table 2. RACI Matrix](#)

2.3 Project Execution Monitoring

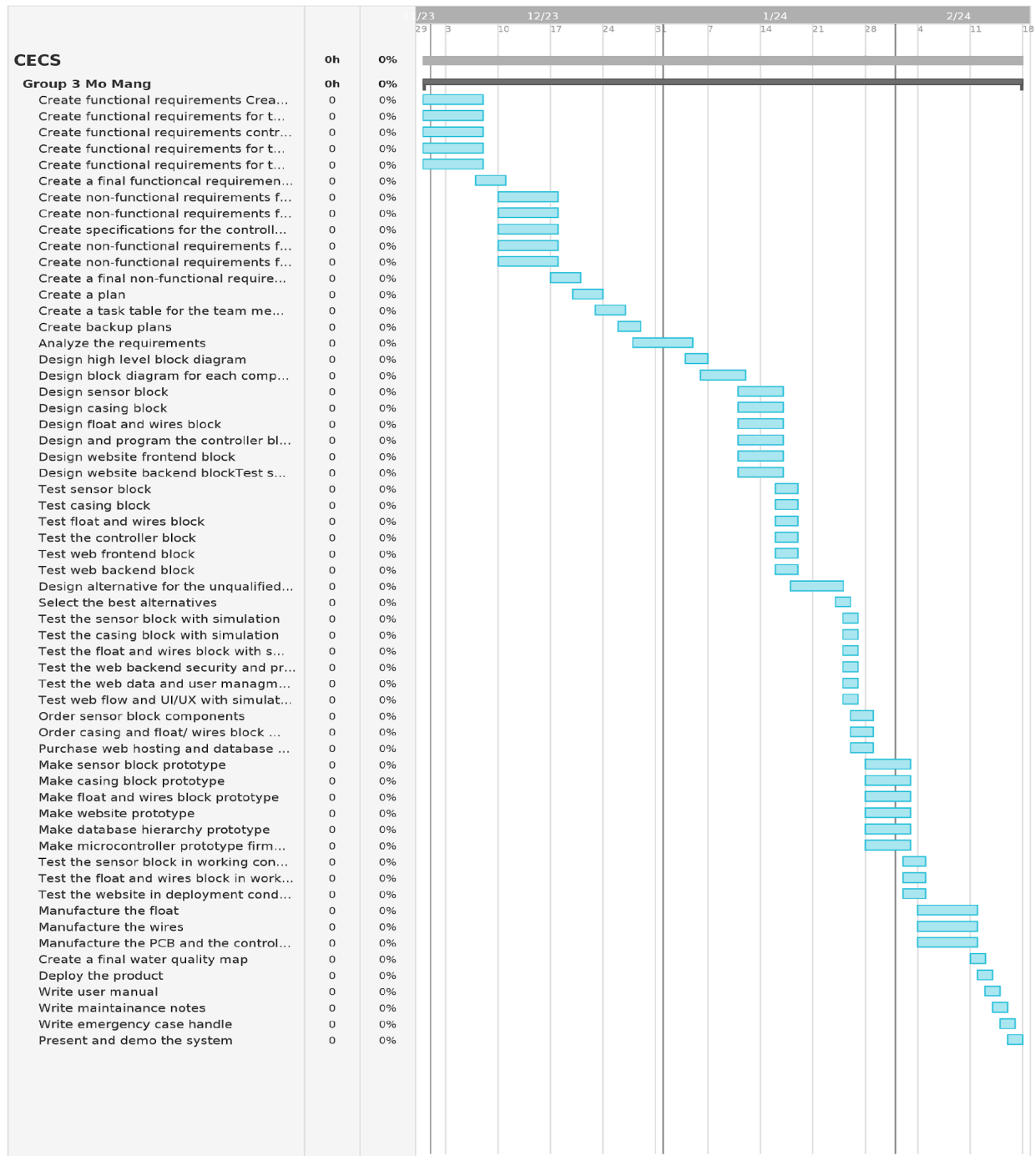


Table 3. Gantt chart

2.4 Challenges and Decision Making

2.4.1 Challenges:

Sensors and calibration:

On the website of the manufacturer, all of the sensor calibration was done in 5v logic while our chosen microcontroller can only operate at 3.3v logic which makes all of the pre-calibration figures and library from the manufacturer unsuitable for our team to use. One spectacular case is the TDS sensor which only works with 5v logic. The readings from the sensors, while quite stable, are a bit noisy. The necessary stats of the sensors like response time and accuracy is not supplied by the manufacturer.

Containers and float:

The whole container is intended to be dipped into the water body and connected to float on the top, this forces the container to be waterproof otherwise all of the electrical components will become useless. The float was hard to choose as we had to ensure it was small enough while having enough buoyancy.

Database and graphs:

The microcontroller will try to plot data in real time, this means a suitable database for logging real-time data while maintaining high data security is needed. Also, we try to minimize the data delivery delay time so a friendly database with a simple API and high configuration is preferred. We also have to plot data in real time without relying on our own computational power as we will not log data into our local machine so we have to find an online graph service provider.

Device portability and configuration:

We want our device to be highly configurable whether it is working in a wifi ready environment or not. We also want it to be highly flexible in changing connectivity settings to fit a wide range of working condition

Power Consumption and API limitation:

Since the device runs entirely on battery, we have to implement many power-saving techniques to save as many runtimes as possible, also there is a limit to how many calls to the server we can make in a few second to ensure safety against DDOS attack

Website UI and UX:

The monitoring website has to have a modern look with simplicity and visible data for observation

Microcontroller memory arrangement:

The microcontroller ran out of memory while adding new feature

2.4.2 Decision Making:

Sensors and calibration:

We have to conduct our own experiment for calibration and extracting more information from each sensor. For sensors that require a 5v logic to operate, we add a voltage divider circuit between the signal line and the ground to shift the signal logic level back to 3.3v. We also implement a median filter for noise reduction.

Container and float:

We have used many waterproof materials and components like rubber rings, silicon glue, watertight box to completely block any water from pouring in, for the float, we decided that for now, we will remove it to reduce cost, increase mobility and ease of installation

Database and graphs:

We decided to use google spreadsheet as our database since it is extremely easy to use and easy to set up, for graphing, we use ThingSpeak for the realtime plot as it is easy to call and easy to integrate into our existing website

Device portability and configuration:

To enable our device to work in different area with different wifi setting, we have implement a wifimanager code onto the chip so every time a wifi configuration is changed, the user can change wifi configuration information on the chip using their phones. We also utilize an offline working mode with restrain features.

Power consumption and API limitation:

We have implemented a 20-second interval between each data uploading time for testing. In operation, we intended to increase this number up to 10 minutes or 30 minutes between each reading to ensure the API call and the power consumption lie within the allowed range. Also, we put the microcontroller to sleep and turned on the LCD screen backlight between data updates to save even more power

Website UI/UX:

We decided to make use of Google Map API with its elegant and straightforward map and marker design we also implemented component like pop up and reduce any residue to ensure maximum simplicity

Microcontroller memory arrangement:

By arranging the memory: reducing the memory space for OTA update and increasing memory space for operation, we have effectively double the amount of memory for adding new feature

Chapter 3: System Design

3.1 Block diagram of the system

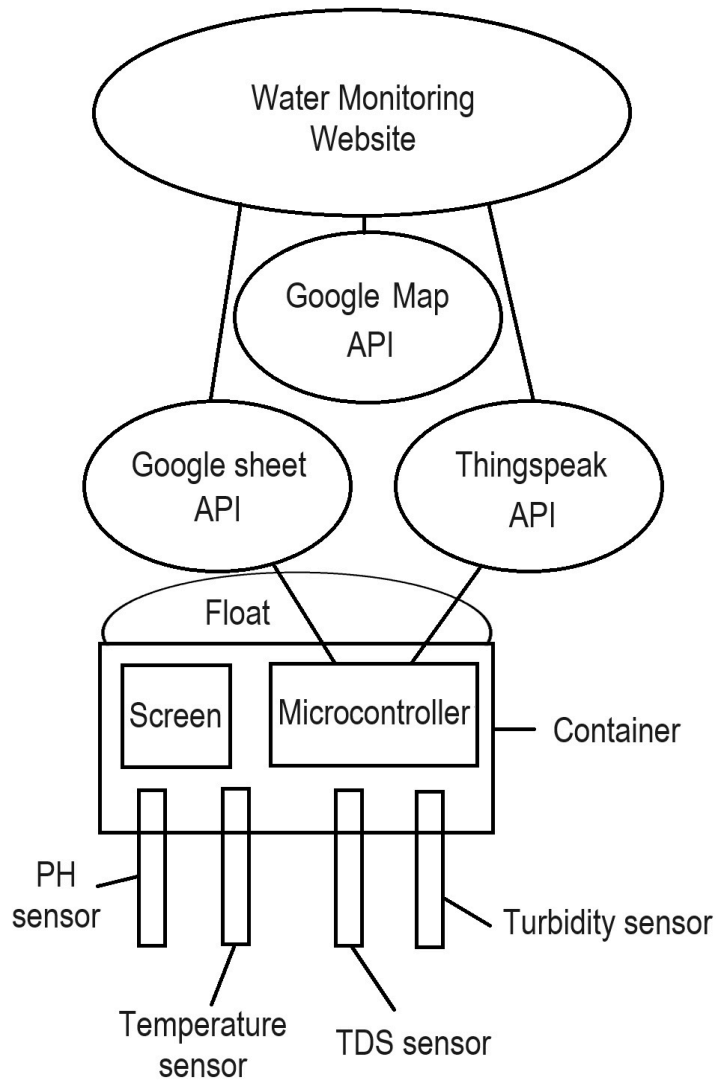
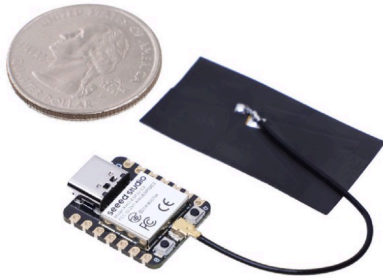


Fig 1. Block diagram of the system

3.2 Design of each block and select the best alternative

Sensor Block:



Seed Studio XIAO ESP32C3 RISC-V With Wi-Fi And Bluetooth 5.0	Power Bank
Dimensions: 21 x 17.5mm	Material: Aluminum alloy + ABS + V0 flame retardant
Circuit operating voltage: 3.3V @ 200mA	Battery type: Lithium Polymer
Charging current: 50mA/100mA	Capacity: 20000mAh
Input voltage (VIN): 5V	Input (Type-C1): 12V-1.5A
Deep Sleep Power Consumption:	Output (USB): 12V-1.5A
Deep Sleep Mode: >44 μ A	Total output current: DC5V-3A
Wi-Fi Enabled Power Consumption:	Dimensions: 20000mAh: 1446626.6mm
Active Mode: <75 mA	Certifications: FCC CE ROHS
Modem-sleep Mode: <25 mA	
Light-sleep Mode: <4 mA	



Wire Digital Temperature Probe DS18B20 with 1m Wire

Operating voltage: 3.3 - 5.5VDC

Current consumption: 11.5mA

Communication standard: Digital TTL 1-Wire (with only one Data pin; note that the Data pin of the sensor needs to be pulled up to VCC level through a 4.7k Ohm or 10k Ohm resistor before connecting to the microcontroller).

Measurable temperature range: -55-125°C

Accuracy: $\pm 0.5^{\circ}\text{C}$

Resolution: 912 bits (adjustable)

Response time < 750ms

Designed as a 1m long waterproof wire with a sturdy steel protective casing.

DFRobot Gravity Analog pH Sensor/Meter Kit For Arduino

SKU: SEN0161

Model: DFRobot Gravity Analog pH Sensor/Meter Kit For Arduino

Module Power: 3.3 - 5VDC

Module Size: 43 x 32mm (1.69x1.26")

Measuring Range: 0-14 pH

Measuring Temperature: 0-60 °C

Accuracy: ± 0.1 pH (25 °C)

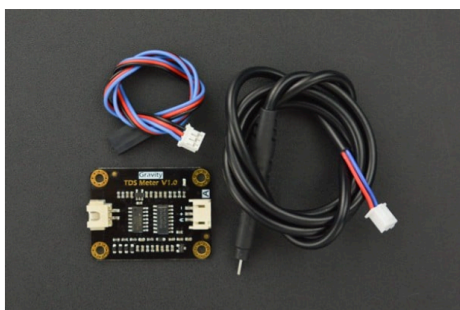
Response Time: ≤ 1 min

pH Sensor with BNC Connector

pH 2.0 Interface (3-foot patch)

Gain Adjustment Potentiometer

Power Indicator LED



DFRobot Gravity Analog TDS Sensor/Meter For Arduino SKU: SEN0244 Model: DFRobot Gravity Analog TDS Sensor/Meter for Arduino Signal Transmitter Board Input Voltage: 5 - 5.5V Output Voltage: 0 - 5V Working Current: 3 ~ 6mA TDS Measurement Range: 0 ~ 1000ppm TDS Measurement Accuracy: $\pm 10\%$ F.S. (25 °C) Module Size: 42 * 32mm Module Interface: PH2.0-3P Electrode Interface: XH2.54-2P TDS probe Number of Needles: 2 Total Length: 83cm Connection Interface: XH2.54-2P Color: Black	DFRobot Gravity Analog Turbidity Sensor For Arduino SKU: SEN0189 Model: DFRobot Gravity Analog Turbidity Sensor For Arduino Operating Voltage: 3.3 - 5V DC Operating Current: 40mA (MAX) Response Time: <500ms Insulation Resistance: 100M (Min) Output Method: Analog Analog Output: 0- 3.3v Digital Output: High/Low level signal (threshold value adjustable via potentiometer) Operating Temperature: 5°C90 °C Storage Temperature: -10°C90°C Weight: 30g Adapter Dimensions: 38mm28mm10mm/1.5inches 1.1inches0.4inches
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3.3 Testing of each block

3.3.1. Sensor Block:

From our experiment we have concluded that:

- Google Spreadsheet API delay time: 300ms
- ThingSpeak API delay time: 20s
- Temperature Accuracy: +0.36/-0.36
- TDS Accuracy: +91/-91
- PH Accuracy: +0.6/-0.6
- Turbidity Accuracy: +3/-3
- On Battery Runtime: 2 days 9 hours 24 minutes
- Temperature sensor response time: 2 minutes
- PH sensor response time: 2 minutes
- TDS sensor response time: 20 seconds
- Turbidity sensor response time: 20 seconds
- PH sensor calibration equation: $2.9545x^3 - 10.3578x^2 + 20.7283x - 9.3570$
- TDS sensor calibration equation: $345230x^2 - 60700x + 1120$
- Temperature sensor calibration equation: $-0.00022x^2 + 1.0842x + 1.75$
- Turbidity sensor calibration equation: $-1120.4x^2 + 5742.3x - 4352.9$
- Sensor Calibration Graphs:

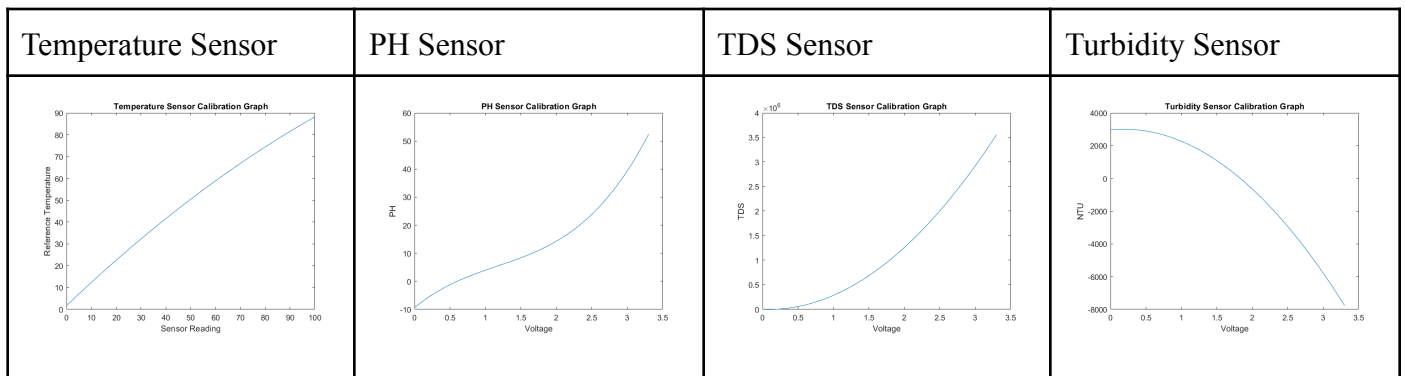
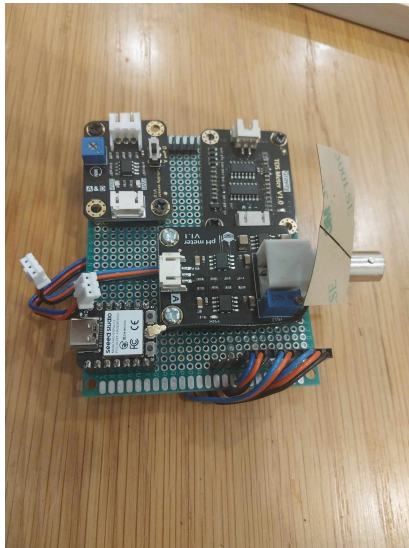


Fig 4. Sensor calibration graph

3.4 Manufacturing and assembly (Implementation)

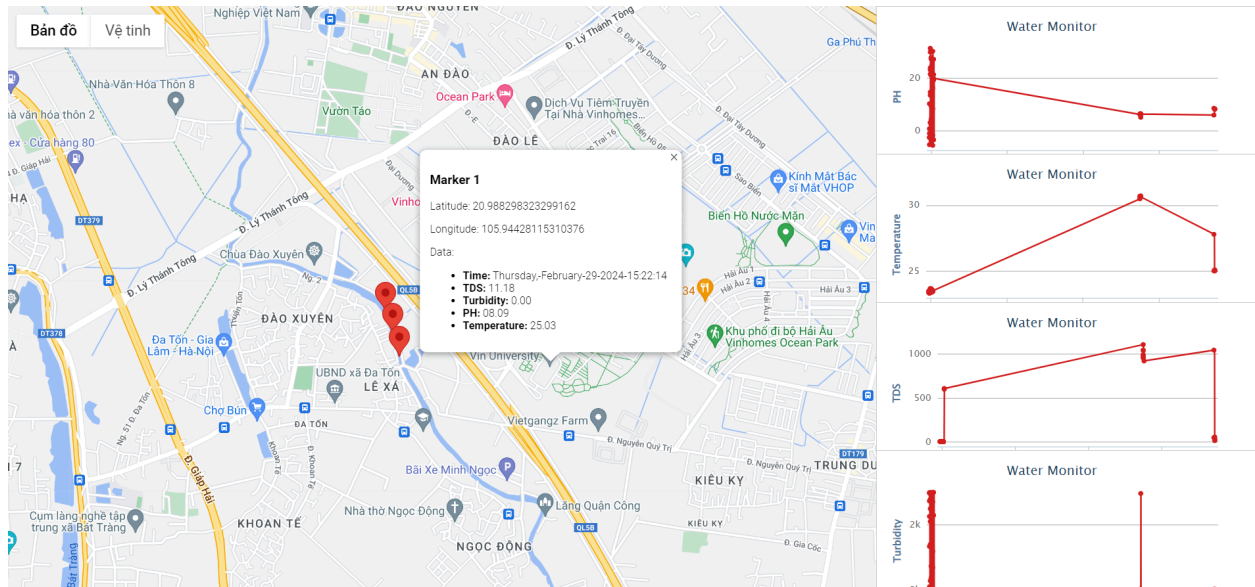
Sensor Block:



Web Display:

	A	B	C	D	E
1	Time	TDS	Turbidity	PH	Temperature
2	Monday,~January-22-2024-12:09:56	1.93	0	1.54	25.03
3	Monday,~January-22-2024-12:10:07	2.21	0	1.55	24.96
4	Monday,~January-22-2024-12:10:18	2.34	0	1.55	24.89
5	Monday,~January-22-2024-12:10:51	1.72	0	1.54	24.82
6	Monday,~January-22-2024-12:11:03	2.17	0	1.55	24.82
7	Monday,~January-22-2024-12:11:14	2.21	0	1.55	24.82
8	Monday,~January-22-2024-12:11:32	1.9	0	1.54	24.89
9	Monday,~January-22-2024-12:11:44	2.07	0	1.54	24.89
10	Monday,~January-22-2024-12:11:56	2.24	0	1.54	24.89
11	Monday,~January-22-2024-06:42:06	1.93	0	1.55	24.96
12	Monday,~January-22-2024-06:42:17	2.03	0	1.55	24.96
13	Monday,~January-22-2024-06:42:29	2.31	0	1.55	25.03
14	Monday,~January-22-2024-06:42:45	2.1	0	1.54	24.96
15	Monday,~January-22-2024-06:42:57	2.24	0	1.55	24.96
16	Monday,~January-22-2024-06:43:08	2.41	0	1.55	24.96
17	Monday,~January-22-2024-06:43:19	2.17	0	1.55	24.96
18	Monday,~January-22-2024-06:43:30	2.45	0	1.55	25.03
19	Monday,~January-22-2024-06:43:42	2.28	0	1.55	25.03
20	Monday,~January-22-2024-06:43:53	2.31	0	1.55	24.96
21	Monday,~January-22-2024-06:44:04	2.21	0	1.55	25.03
22	Monday,~January-22-2024-06:44:16	2.38	0	1.55	24.96
23	Monday,~January-22-2024-06:44:27	2.17	0	1.55	25.03
24	Monday,~January-22-2024-06:44:39	2.55	0	1.55	27.58
25	Monday,~January-22-2024-06:44:50	2.21	0	1.55	32.2

Online Database



Website interface with specific data and visualize graph



Color marking system to determine the pollution level

Chapter 4: System Testing and Analysis

4.1 System Testing

4.1.1. Test the durability of the hardware block

Objective

The target of this test phase is to grade the durability and operational in real life of the hardware component (all of the sensors) in our Water Monitoring Live Map system. It is important that the sensor block preform accurately, and smoothly and can withstand harsh environments such as underwater for a long time.

Test Procedure

The hardware component will have to go through a durability test, it will be fully submerged in a water environment for one week. First, it will not carry any electrical circuit, and then we implemented the circuit in the second phase of the test. The test will simulate real-world conditions where the hardware be drowned into water, ensuring the operational still going even after a long time.

1. Preparation:

- Ensure the hardware is configured as it would be during standard conditions, including all necessary power sources and electrical circuits.
- The water used for testing will be maintained at a standard temperature, clear natural water , approximately 25°C (77°F), to replicate typical river conditions.

2. Submersion:

- The hardware will be placed in a transparent, water-filled containment unit to allow checking during the test duration.
- The hardware will be fully submerged, ensuring that water covers all parts are covered in water.

3. Monitoring and Data Collection:

- Inspections will be conducted daily to check for any signs of water leaks or physical damage.
- The hardware's operational ability will be tested remotely by checking the set data at the database. This includes data transmission, sensor accuracy, and power efficiency.

4. Post-Test Evaluation:

- When one-week submersion is complete, the hardware will be removed from the water and look for any signs of water damage.
- A out-of-water test will be conduct to test all sensors in the lab environments
- The hardware will be disassembled and reassemble after the lab experiment.

Expected Outcomes

- The hardware should be completely dry, no sign of water
- The sensors have to work normally during the test and after the test
- The sealed/waterproof material should not be damaged

Conclusion

The test is going to check the durability of the waterproof container, from there we can see the ability to protect all the inside components of the container and make suitable adjust for improvement.

4.1.2. Test the power supplier

Objective

The test will mark the ability to power the system of the power bank to see how long it can last in a prototype in a working environment. This test is important to ensure the ability to transmit the data for a long period without mistakes and also ensure the optimization code works smoothly within the hardware component.

Test Procedure

The power supply will be involved a simulated run, designed to replicate the power consumption of the system. We will record the working time and the accuracy of the system and based on that to judge the power bank.

Preparation:

Set up a simulation with the same power consumption as the system

Set demo data with a fixed standard to compare in order to measure the accuracy of the system

Simulation Environment:

With the built-in power supply module of the power bank, we can ensure a suitable level of input voltage for the system to operate.

Continuous Operation:

Monitor the power supply continuously and record the battery level regularly to track consumption rates and identify any error or voltage drop, power drains or inefficiencies.

Post-Test Analysis:

Identify any errors, and judge whether the system will be able to work in that same condition for a long period. Analyze the battery life span to adjust further.

Find an alternative source of power if the old battery cannot meet the expectations.

Expected Outcomes

The battery should power the prototype at a normal working rate in the whole testing phase

There should be no voltage drop, error or power drain

Conclusion

This power supply durability test is important in ensuring our system remains operational and reliable in the field over many months operating. By evaluating the power supply's use of time and efficiency, we aim to provide a product that minimizes maintenance requirements and maximizes user convenience and that can support accurate water quality monitoring.

4.1.3. Test the Operation

This is one of the most important steps for operating our system. First of all, we need to connect to the Seeed Studio XIAO ESP32C3 RISC-V and evaluate the connection quality. In some cases, we even could not access the WIFI or it would automatically disconnect. Therefore, carefully checking this step is essential for risk management when implementing. Then, we run the keyboard control example using C++. If the system performs smoothly and reports to the database adaptively in real-time without any unexpected errors, it is ready for the next phase.

4.1.4. Overall test

After finishing the initial test for each component, we combine them all and begin to upload the sample code for the real implementation on the system. However, further steps still exist to make this system run completely smoothly. These phases will be mentioned in the next part.

4.2 Results, Analysis, and Discussion

4.2.1. Original Algorithm

Data filter code

To increase accuracy and smoothness of data reading we also implement a median filter code

```

float getVoltageFilterAvg(uint8_t sample, uint8_t sensorPin) // median filter
{
    uint8_t i;
    float val_buff[sample];
    float val = 0.0;

    for (i = 0; i < sample; i++)
    {
        val_buff[i] = analogRead(sensorPin);
        delayMicroseconds(10);
    }
    /* Sort buffer */
    selectionSort(val_buff, sample);
    /* get averages in the middle of the buffer */
    for (i = sample / 4; i < (3 * (sample / 4)); i++)
        val += val_buff[i];
    val = val / (sample / 2);
    val = (val / 4096) * 3.3;
    return val;
}

```

Microcontroller Wifi configuration and API calls:

To ensure device portability and working in different conditions, we have implemented a wifi configuration code via wifimanager library and write code for both offline and online working operation.

```

void configWifi()
{
    if (WiFi.waitForConnectResult() != WL_CONNECTED)
    {
        lcd.setCursor(0, 0);
        lcd.print("Connecting...");
        WiFiManager wifiManager;
        wifiManager.resetSettings();
        wifiManager.setConfigPortalTimeout(30);
        wifiManager.autoConnect("WaterSensor01");
    }
}

```

Sensor measurement reading:

To read data from the sensors effectively, we send a parameter about the reading pin and the size of the sliding window for the median filter algorithm, after going through the filter, the data will then be calibrated by the calibration equation we have found by experimenting earlier

```

float PH = GetPHCalibrated(getVoltageFilterAvg(20, 4));
float TDS = GetTDSCalibrated(getVoltageFilterAvg(20, 5));
float Turb = GetTurbCalibrated(getVoltageFilterAvg(20, 3));
float Temp = GetTempCCalibrated(ds.getTempC());

```

Spreadsheet and ThingSpeak API:

To send data to the Google Spreadsheet Database and ThingSpeak for information storing and graphing, we utilize SpreadSheet API and ThingSpeak API in our code

```
valueRange.add("majorDimension", "COLUMNS");
valueRange.set("values/[0]/[0]", asString);
valueRange.set("values/[1]/[0]", String(TDS));
valueRange.set("values/[2]/[0]", String(Turb));
valueRange.set("values/[3]/[0]", String(PH));
valueRange.set("values/[4]/[0]", String(Temp));

success = GSheet.values.append(&response, spsID, "Sheet1!A1", &valueRange);
```

```
ThingSpeak.setField(1, PH);
ThingSpeak.setField(2, Temp);
ThingSpeak.setField(3, TDS);
ThingSpeak.setField(4, Turb);
int x = ThingSpeak.writeFields(myChannelNumber, myWriteAPIKey);
```

Error Catching:

The following code catch error in case there is a corruption or interruption in data sending.

```
if (x == 200)
{
  Serial.println("Channel update successful.");
}
else
{
  Serial.println("Problem updating channel. HTTP error code " + String(x));
}
delay(10000);
}

else
{
  Serial.println(GSheet.errorReason());
  Serial.println();
  if (success)
  {
    Serial.println(ESP.getFreeHeap());
  }
}
```

LCD screen display:

The LCD will turn its backlight off to save power when it's not changing its display

```

lcd.clear();
lcd.backlight();
lcd.setCursor(0, 0);
// print message
lcd.print(temp);
lcd.setCursor(9, 0);
lcd.print(ph);
lcd.setCursor(0, 1);
// print message
lcd.print(tds);
lcd.setCursor(9, 1);
lcd.print(ntu);
delay(3000);
lcd.noBacklight();

```

The website code:

```

const lastRow = data.values[data.values.length - 1];
var shouldChangeColorGreen = parseFloat(lastRow[3]) >= 6 && parseFloat(lastRow[3]) <= 9
&& parseFloat(lastRow[2]) >= 0 && parseFloat(lastRow[2]) <= 100
&& parseFloat(lastRow[4]) >= 20 && parseFloat(lastRow[4]) <= 30
&& parseFloat(lastRow[1]) >= 0 && parseFloat(lastRow[1]) <= 1000;

```

```

var shouldChangeColorRed = parseFloat(lastRow[3]) < 6 && parseFloat(lastRow[3]) > 9
&& parseFloat(lastRow[2]) < 0 && parseFloat(lastRow[2]) > 100
&& parseFloat(lastRow[4]) < 20 && parseFloat(lastRow[4]) > 30
&& parseFloat(lastRow[1]) < 0 && parseFloat(lastRow[1]) > 1000;

```

Compare the collected data with government standard

```

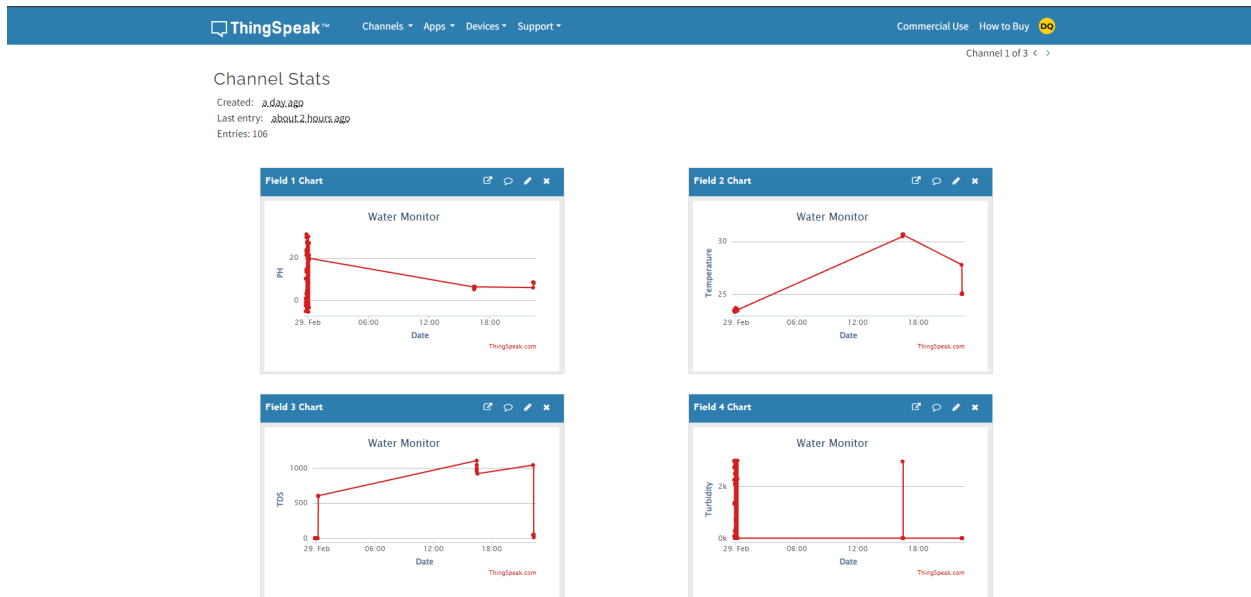
if (shouldChangeColorGreen) {
  marker.setIcon('http://maps.google.com/mapfiles/ms/icons/green-dot.png');
}else if(shouldChangeColorRed) {
  marker.setIcon('http://maps.google.com/mapfiles/ms/icons/red-dot.png');
}else{
  marker.setIcon('http://maps.google.com/mapfiles/ms/icons/yellow-dot.png');
}
openChartPanel(location, index, lastRow);
})
.catch(error => {
  console.error('Error fetching data:', error);
});

```

Change the color of the marker due to the comparison result. If no criteria is over the limitation, set the marker green. If no criteria is over the limitation, set the marker red. Others are yellow.

4.2.2. Problem Solutions

Problem: Data is not visualized adequately, hindering effective analysis and interpretation.



→ **Solution:** Using ThingSpeak, a data visualization tool, we can create visually graphs and charts. By presenting the data visually, users can easily identify the data about the water quality. Visualizing the data also give user a better look on the trend and the lately quality of that water source.

Problem: We want a quickway for the user to know whether the water body is clean or not, need something that directly connect to the data but not the data it self because reading take a lot of time.

→ **Solution:** Using the color marking system, we can tell the health of that water source. If the marker is green then all of the stars are above average and bellow the limit. If it is yellow, it means some of the data are in a danger zone and might be harmful for the user. If it is red then all of the statistics is above the limit and is extremely dangerous for human use.

Chapter 5: Conclusion and Recommendation

5.1 Conclusion

The importance of advanced water quality monitoring solutions in our time is undeniable and their integration into public health and environmental management structures should be widely applied. The need to improve water monitoring technologies is critical to show humanity to a horizon of ecological balance and long-term well-being. This journey not only helped us discover a new horizon of new knowledge, but also gave us a deep understanding of environmental science, data analysis, and the a lot of complex mechanics of real-time monitoring systems.

Team collaboration:

From the beginning of the project, our team recognized the importance of collaboration. It consisted of clear communication and respect for others. Using tools like the RACI Matrix has greatly enhanced our task management and accountability capabilities, allowing us to navigate project complexities with flexibility and precision. Our diverse backgrounds and skill sets have enabled us to approach challenges from many angles and create comprehensive solutions.

Time management and organization:

The list of tasks and deadlines inherent in a project of this magnitude underscores the importance of time management. By identifying important milestones and evaluating task priorities, we completed the project carefully. We have allocated our work and break times effectively to ensure that our productivity and focus remain at peak levels. The end result is a project that is not only on schedule but also of outstanding quality.

Hardware related skills:

Through this project, we gained knowledge in many technical fields, from hydrology and environmental science to IoT and data analytics. The hands-on experience of deploying our hardware in underwater environments gave us the opportunity to apply theory learned from lecture. This is a tool that helps us improve our technical skills, especially in areas like sensor integration, data encryption, and real-time analysis.

Software related skills:

Our Water Monitoring Live Map is a software framework designed to synthesize, process, and visualize water quality data streams. The development process was fraught with problem-solving and debugging, requiring a deep dive into algorithm optimization, system structures, and user interface design. Building software for water systems requires high precision and attention to detail. Many tests and debugging have been performed to ensure that the model works correctly and without errors.

Soft-skills:

To address the challenges through the project, we have addressed technical barriers to ensure hardware operability across diverse aquatic environments. The process of design, testing and debugging has provided us with an end product of the highest quality finish.

In short, this project was a challenge of learning, adapting and collaboration. It has pushed us through many experiences, from the thrill of breakthroughs to the seriousness of problem solving. As we reflect after the process, we believe that the knowledge and skills gained through this will be the important steps for our future success, both academically and professionally.

5.2 Future Recommendations

Throughout this project, we encountered challenges mainly related to the integration of the Seeed Studio XIAO ESP32C3 RISC-V microcontroller with the power module, sensor accuracy as well as system performance. power supply system. Based on these experiences, we propose the following recommendations to address these issues in future project iterations:

- ***Advanced microcontroller and power module integration:***
 - Choose other microcontroller models with improved compatibility and stronger support for power management features to improve the stability of the connection between the microcontroller and the power part.
 - Use advanced power management protocols in the microcontroller firmware to optimize power consumption, from that extend the device's use life.
- ***Sensor quality and calibration:***
 - Identify higher-end sensors with a proven track record of accuracy and durability in aquatic environments. Prioritize sensors with industry standard certification and extensive validation studies.
 - Establish a regular calibration process for your sensors, leveraging both laboratory and field calibration methods to ensure consistent data accuracy over time. Consider integrating automatic calibration functions within the system to minimize manual intervention.
 - Explore the possibility of developing a multi-sensor fusion method, where data from multiple sensors is combined and analyzed using advanced algorithms to improve overall measurement accuracy and reliability.
- ***Power supply efficiency and sustainability:***
 - Consider other battery technologies which can provide higher energy density, faster charging and longer life cycle. Some options can be lithium-sulfur or solid-state batteries.
 - Design a more adaptive energy management system capable of dynamically adjusting energy consumption.
 - Balance operating with energy conservation principles to maximize efficiency.
- ***Integrated positioning and mapping:***
 - Implement geolocation services to automatically detect a user's current location or allow users to pinpoint the exact location of their water quality concerns on an

interactive map. This ensures accurate data collection and facilitates identification of potential problem areas.

- Allows users to view existing reports in their neighborhood to avoid duplicate submissions and provide a broader understanding of local water quality issues.
- ***Detailed report form:***
 - Design a reporting form to collect data, including the type of concern (e.g., discoloration, odor, contamination).
 - Provide options for users to upload visual evidence to support their reports.
- ***Anonymous reporting:***
 - Submit reports anonymously to encourage people concerned about privacy to report. However, anonymous reports can still be verified and tracked, with aim of maintaining data integrity.
- ***Feedback and follow-up mechanism:***
 - Establish a system for acknowledging received reports and providing feedback on the actions taken or planned. Update the status of reported issues. This can foster trust and encourage ongoing community participation.
 - Add a feature for users to subscribe to updates on specific reports or general water quality issues in their area, which can keep the community engaged and informed.

By addressing these key points, we can enhance the performance, reliability, and sustainability of the Water Monitoring Live Map. These improvements will not only improve the user experience but also contribute to the broader objective of promoting environmental protection and public health.